PANKHURI KULSHRESTHA

MACHINE LEARNING RESEARCH ENGINEER

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Profile Summary

Dedicated and highly skilled Machine Learning Research Engineer with a strong background in computer science, mathematics, and statistics. Proficient in developing innovative machine learning algorithms and models to solve complex problems. Experienced in handling large datasets and implementing data preprocessing techniques to extract meaningful insights. Adept at utilizing various machine learning frameworks and libraries to design and deploy scalable and efficient solutions.

Research Interests

- Computer Vision
- SLAM
- Generative AI
- Human-AI Interaction

Education

MSc Artificial Intelligence *16 Jan 2022 - 20 Jan 2023*

University of Essex, UK

Bachelor of Engineering *31 May 2007 - 31 May 2011*

Rajiv Gandhi Proudyogiki Vishwavidyalaya, India

Specific Area of Work

- 3D Deep Learning (PointNet, SPVCNN)
- 2D Deep Learning (Vision Transformers, CNN)
- Edge AI
- GAN

Technical Skillset

- Languages: Python, C,C++, Linux Shell Scripting, HTML
- Frameworks/Libraries: Keras, Pytorch, TensorFlow, ONNX, OpenCV, Scikit-learn
- Cloud Technologies: AWS, Kubernetes, GCP, Docker
- Platforms: Ubuntu Linux 12.04, 16.04, 20.04; Windows XP, 7, 8, 10

Achievements & Certifications

- Earned Google Developer Badge in Machine Learning Bootcamp , Oct 2022
- Tensorflow Certification, Oct 2022
- Ranked within 14% Hub map Organ Segmentation Kaggle competition, 22 Sept 2022
- Coursera Deep Learning Specialization, 15 Aug 2022
- Post Graduate Program in Artificial Intelligence and Machine Learning, 31 Dec 2018 – 31 Dec 2019

Work Experience

Machine Learning Research Engineer 16 Jan 2023-Till Date

Mindtrace, Manchester

- Worked towards Industrial Automation using advanced computer vision and 3D deep learning
- Fine-tune DINO v2 models using LoRA (Low-Rank Adaptation) for weld defect classification
- Currently working on Point Cloud data and 3D deep learning Image segmentation models like PointNet and SPVCNN.
- Gained experience in Geospatial libraries like PDAL, GeoPandas, Shapely for preprocessing LIDAR data and building performance efficient algorithms in Python and C++ to handle heavy 3D data processing.
- Worked to infer and compare performance of Yolo v3 model on Kneron kl520 USB AI dongle.
- Worked on deploying models in AWS and GCP and model inference using Batch Jobs.

Computer Vision Engineer 1 Jun 2021 – 16 Dec 2021

Edcast, India

- Worked on Image Processing using OpenCV and Object detection for desktop and mobile applications
- Implemented Text detection and extraction using Google Vision API and Tesseract OCR
- Utilized FLANN based Matcher and SIFT algorithms for object identification
- Worked on AWS Services like Lambda/API Gateway/S3 with serverless framework.

Software Engineer 29 Sep 2019 – 31 May 2021

IQVIA, India

- Worked on a computer-implemented method for classifying and interpreting life science documents for a Content Management System. The solution designed for classifying documents involved text analysis of digitized representations to identify raw words in the text, construct analysis to identify document elements and relative spatial positioning of document elements on a page, image analysis to identify images and process the identified images to extract text.
- Worked on Document Layout Analysis models comparing Faster RCNN and Mask RCNN
- Developed pipelines for text Data Cleansing, feature extraction, model development and deployment using Gitlab, Kubernetes and Dockers.
- Created Kubernetes job to deploy Machine Learning models inside Kubernetes cluster and mount Persistent Volumes using Helm.

Software Engineer 24 Sep 2017 – 22 Sep 2019

Bank of Paribas, India

- Worked on enabling deployment process for Bank Applications to use Continuous Integration and Continuous Delivery Platform using Jenkins and TeamCity.

ACADEMIC PROJECTS

Chatbots

Sept 2022 – Dec 2022

The aim of the project was to analyze the role of Attention based Neural networks for dialog modelling of Conversational AI Agents. The chatbot primarily is expected to understand the User intention and determine the type of response to provide relevant information or follow up questions exhibiting a human like behavior. A Natural Language Understanding Component is developed to parse the conversation to identify the User Intention and extract the associated information. A Reformer model, also known as the efficient Transformer, is used to generate a dialogue between two bots and understand the context of each one. The bot is expected to learn how to answer questions, additionally, will also know how to ask questions if it needs more information. I used this concept to automate customer service for hotel receptions and develop a closed domain chatbot.

Tools : Tensorflow, Trax, BERT, Transformers

IMAGE SEGMENTATION

Aug 2022 – Sept 2022

Worked on Kaggle competition 'HuBMAP + HPA - Hacking the Human Body' to identify and segment functional tissue units (FTUs) across five human organs. The model based on a U-shape network (UneXt50) with EfficientNet as backbone was trained using Lovász and dice loss on TPU and stored on Google cloud storage. Based on a Unet architecture where the encoder part creates a representation of features at different levels, while the decoder combines the features and generates a prediction as a segmentation mask. Additionally, it used Feature Pyramid Network (FPN) to improve model performance and convergence speed. The Atrous Spatial Pyramid Pooling (ASPP) block was added between encoder and decoder to overcome the flaw of traditional U-shape networks resulting by a small receptive field.

Tools : Pytorch, Tensorflow, Python, GPU and TPU Training, Google Cloud Storage

Federated Learning - Human Activity Recognition

April 2022 – June 2022

The aim of the project was to develop a mobile application based on Human Activity Recognition using energy-efficient Internet of Things (IoT) platform. The mobile application enabled the user to identify, track and monitor eleven activities using their Smartphones and wearable devices. I integrated Empatica E4 wearable device accelerometer, gyroscope, PPG sensors with in-built Android application. The sensor data was collected and stored in SQLite database within Edge device for Training and Inference using Tensorflow Lite. To mitigate the challenge of User data security and privacy, a Federated Learning approach was utilized to train the models.
For Federated Learning,
I evaluated Federated Average, FedProx and Personalized algorithms. I compared and analyzed the accuracy of different model learning techniques like Population, Personalized and Randomized models. I was able to develop an efficient and reliable communication interface for Federated Learning using Message Queue and MQTT protocol. I compared my model results with existing research work and achieved comparable results with combination of Convolution and LSTM layers.

Tools : Tensorflow Lite, Python, Android, Gitlab, Rabbit MQ, JIRA

NL2SQL

Jan 2022 – April 2022

Automated Text to SQL Generation is an automated task which targets to convert Natural Questions into Structured Query Language. These queries are capable to retrieve data stored in Relational Database Management systems easily and efficiently. Writing SQL statements can be tricky as it requires a background knowledge about the SQL syntax and database schemas. In this project, I built Sequence to Sequence models with Encoder-Decoder RNN and LSTM layers and Attention layer to capture the context of the sentence and translate them to SQL queries. I evaluated the performance of the model and observed the Sequence to Sequence approach translated English sentences with moderate accuracy. I used Spider dataset for training and evaluating the deep learning model architecture. The Spider dataset is complex as it covers various combinations of SQL components i.e. multiple column selection, nested queries, joins and aggregation functions. It is cross domain as it includes different database schema which motivates generalized model learning.

Tools : Tensorflow, Python, SQL